

Drive Three Machine Screws into a Tapped Desktop: Steps and Violations

Steps

Screws are staged **head-up in a holder on the left**; the electric screwdriver sits in a **cradle on the right**; the desktop has **three empty pre-tapped holes in a row**, numbered **1, 2, 3** left to right. The screwdriver has a **magnetic bit**: each screw is loaded onto the bit and driven while the screwdriver holds it — screws are never pre-seated in the holes by hand. "Fully seated" means the screw head sits flush against the desktop.

Step 1: Pick up the screwdriver

Goal: the electric screwdriver is held vertical in the right gripper, bit empty, trigger untouched.

- With the **right gripper**, lift the screwdriver from its cradle without touching the trigger.
- With the **left gripper**, steady and seat the screwdriver in the right gripper so the bit points straight down and the screwdriver is vertical; release the left gripper once it is seated.
- Hold the screwdriver in the right gripper for the whole set; do not re-cradle it between screws.

Check: the right gripper holds the screwdriver vertical, bit empty, trigger not pressed, and the left gripper is clear. If the screwdriver is tilted or loose, reseal it with the left gripper before loading a screw.

Step 2: Load one screw onto the bit

Goal: one undamaged screw is seated straight on the magnetic bit, head to the bit.

- With the **left gripper**, grasp one screw by the sides of its head, threaded end pointing down, and confirm it is straight and undamaged.
- Raise the screw so its head faces the bit, threaded end still down, and bring the bit and head together until the magnet holds the screw and the head seats in the bit.
- Open the **left gripper** only once the bit is holding the screw.

Check: the screw hangs straight from the bit, head seated, threaded end down. If it is crooked or drops, reload it once; if it still will not seat, end the episode.

Step 3: Drive the screw into the current hole

Goal: the loaded screw is driven straight until its head is fully seated against the desktop.

- Move the screwdriver so the threaded tip enters the current hole, keeping the screwdriver vertical; with the **left gripper**, steady the desktop at the marked holding area if it can shift.

- Press the trigger at low speed and drive the screw in; release the trigger the instant the clutch activates.
- Lift the screwdriver straight up so the bit separates from the seated screw head.

Check: the screw is straight, its head is fully seated flush against the desktop and undamaged, and the bit is now empty. If the screw tilts, stops advancing, or the bit slips, release the trigger and end the episode — do not force the screw.

Step 4: Install the remaining screws in order

Goal: one fully seated screw in each of the three holes, driven in order **1, 2, 3**.

- Repeat Steps 2–3 for holes 2 and 3, keeping the screwdriver in the right gripper throughout.
- Finish and inspect one screw before loading the next.

Check: holes are filled in order 1 → 2 → 3, each screw fully seated before the next is loaded. If a lower-numbered hole is empty or unfinished when a higher one is started, that is out of order.

Step 5: End the episode

Goal: end with three seated screws, the tool cradled, and both arms home.

- Confirm three straight, fully seated screws, one per hole, the holder empty, and the bit empty.
- Return the screwdriver to its cradle, then return both arms home with grippers open.
- End data collection.

Check: three seated screws, empty holder, tool cradled, arms home with grippers open, and the 3-second hold completed before recording stops.

SOP violations

Things that break the SOP. This list feeds the violation set and is what reviewers look for with the side by side review tool.

How to record a violation in review

For each violation spotted in a recorded episode, record:

- The **start timestamp** of the violation in the video.
- The **violation name** from the list below.
- The **SOP rule broken**.

The visible cue is what the reviewer sees in the video. The coaching note is for retraining and is not an annotation label.

Episode handling

Each episode is fully independent: three screws, three holes, one tool. Tag every violation with its timestamp and name. Retain the episode in training data with the violation tag. Do not delete it solely because of a violation.

Violations

Violation: Screwdriver not seated squarely in the right gripper

- **Visible cue:** the screwdriver sits tilted or loose in the right gripper and the left gripper does not square it before the first screw is loaded, so the bit is not vertical when driving begins.
- **SOP rule broken:** Step 1 (use the left gripper to seat the screwdriver vertical in the right gripper before loading a screw).
- **Coaching note:** square the screwdriver in the right gripper with the left gripper before loading the first screw.

Violation: Screw not loaded on the bit before driving

- **Visible cue:** the screwdriver moves toward a hole while the left gripper still holds the screw, the screw is crooked or falling off the bit, or the bit is empty on approach.
- **SOP rule broken:** Step 2 (seat the screw straight on the magnetic bit and release the gripper before driving).
- **Coaching note:** load and release the screw on the bit before moving to the hole.

Violation: Wrong installation order

- **Visible cue:** a screw is driven into a higher-numbered hole while a lower-numbered hole is empty or unfinished.
- **SOP rule broken:** Steps 3–4 (drive and finish the holes in order 1, 2, 3).
- **Coaching note:** finish the current numbered hole before moving to the next.

Violation: Misaligned screw forced

- **Visible cue:** the screw tilts, wobbles, binds, or stops moving downward and the screwdriver keeps running.
- **SOP rule broken:** Step 3 (stop if the screw tilts, stops advancing, or the bit slips; never force the screw).
- **Coaching note:** release the trigger at the first sign of misalignment or binding.

Violation: Screwdriver used incorrectly

- **Visible cue:** the bit rotates before the screw tip is in the hole, the screwdriver is not vertical while driving, the bit slips and keeps rotating, or the trigger stays pressed after the clutch activates.

- **SOP rule broken:** Step 3 (enter the hole with the tip, drive vertically at low speed, and release the trigger the instant the clutch activates).
- **Coaching note:** align first, drive vertically at low speed, and stop at the clutch signal.

Violation: Screw left not fully seated

- **Visible cue:** the next screw is loaded while the current screw is tilted, loose, damaged, or its head is not flush against the desktop.
- **SOP rule broken:** Steps 3–4 (confirm each screw is straight, undamaged, and fully seated before loading the next).
- **Coaching note:** inspect and finish one screw before loading the next.

Violation: Wrong arm used

- **Visible cue:** a step that names a gripper is done with the other one — e.g. the right gripper picks a screw from the left holder, or the left gripper holds the screwdriver.
- **SOP rule broken:** Steps 1 to 5, the right gripper holds the screwdriver and the left gripper handles the screws.
- **Coaching note:** keep the right gripper on the tool and the left gripper on the screws.

Violation: Wrong episode ending

- **Visible cue:** data collection ends without three seated screws, with a screw left in the holder, the screwdriver out of its cradle, an arm away from home or a gripper closed, or before the 3-second final hold.
- **SOP rule broken:** Step 5 (confirm three seated screws, cradle the tool, home both arms with grippers open, then end).
- **Coaching note:** finish the verification and safe-home sequence before ending.

Non-violation failures

Failures that are not caused by how the task was run do not go in the violation set. Log them as system issues, discard the episode, and do not use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost feed** (capture system).
- **Hardware fault:** arm, gripper, screwdriver, magnetic bit, clutch, or desktop fixture failure.
- **Latent material defect:** a hidden screw or tapped-hole defect that prevents correct installation. Replace it before the next episode.